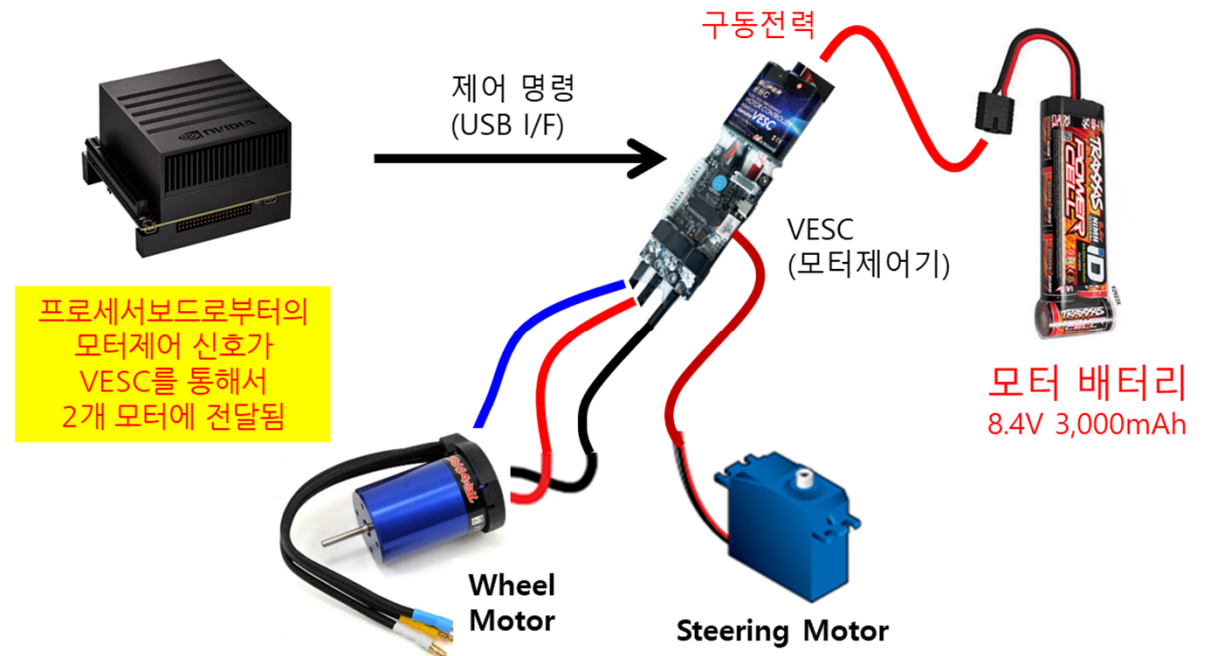


모터제어기 VESC

모터제어기 VESC 설정 방법

모터제어기 - VESC

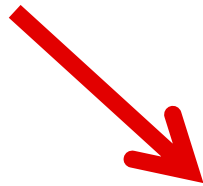
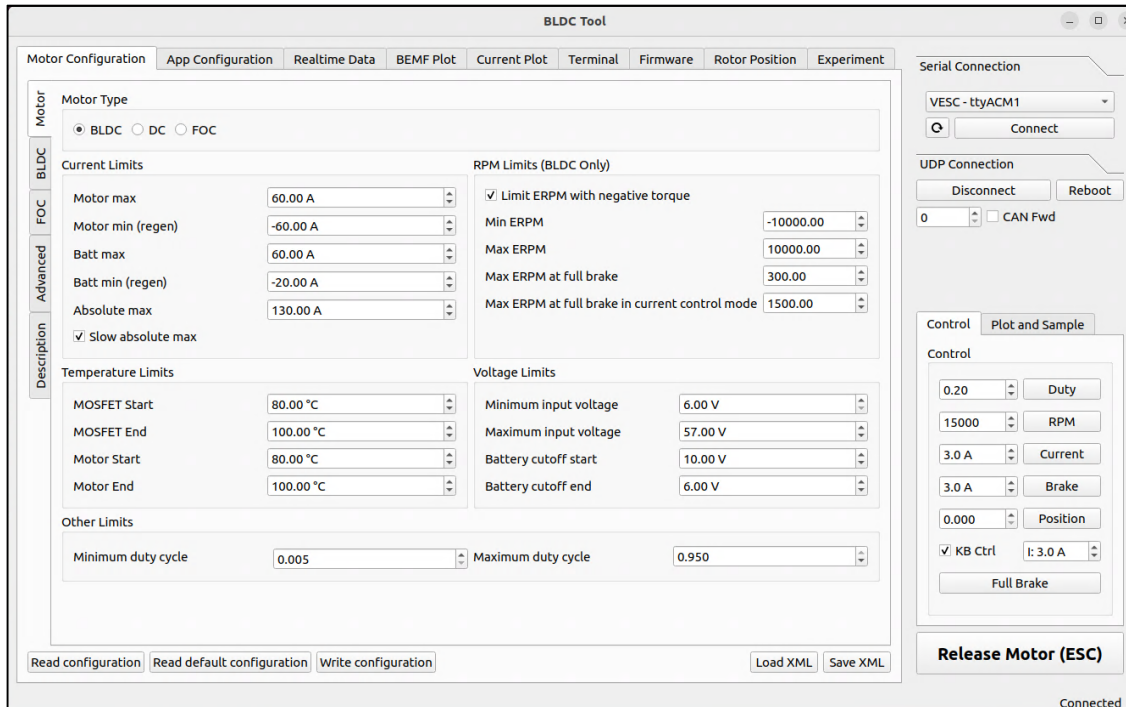
1. 현재 VESC의 펌웨어 버전은 2.18
2. "old_vesc_tool" 사용해서 VESC 설정값을 세팅한다.



VESC 설정값 세팅

- 터미널에서 아래와 같이 입력해서 VESC 설정도구를 실행한다.

old_vesc_tool

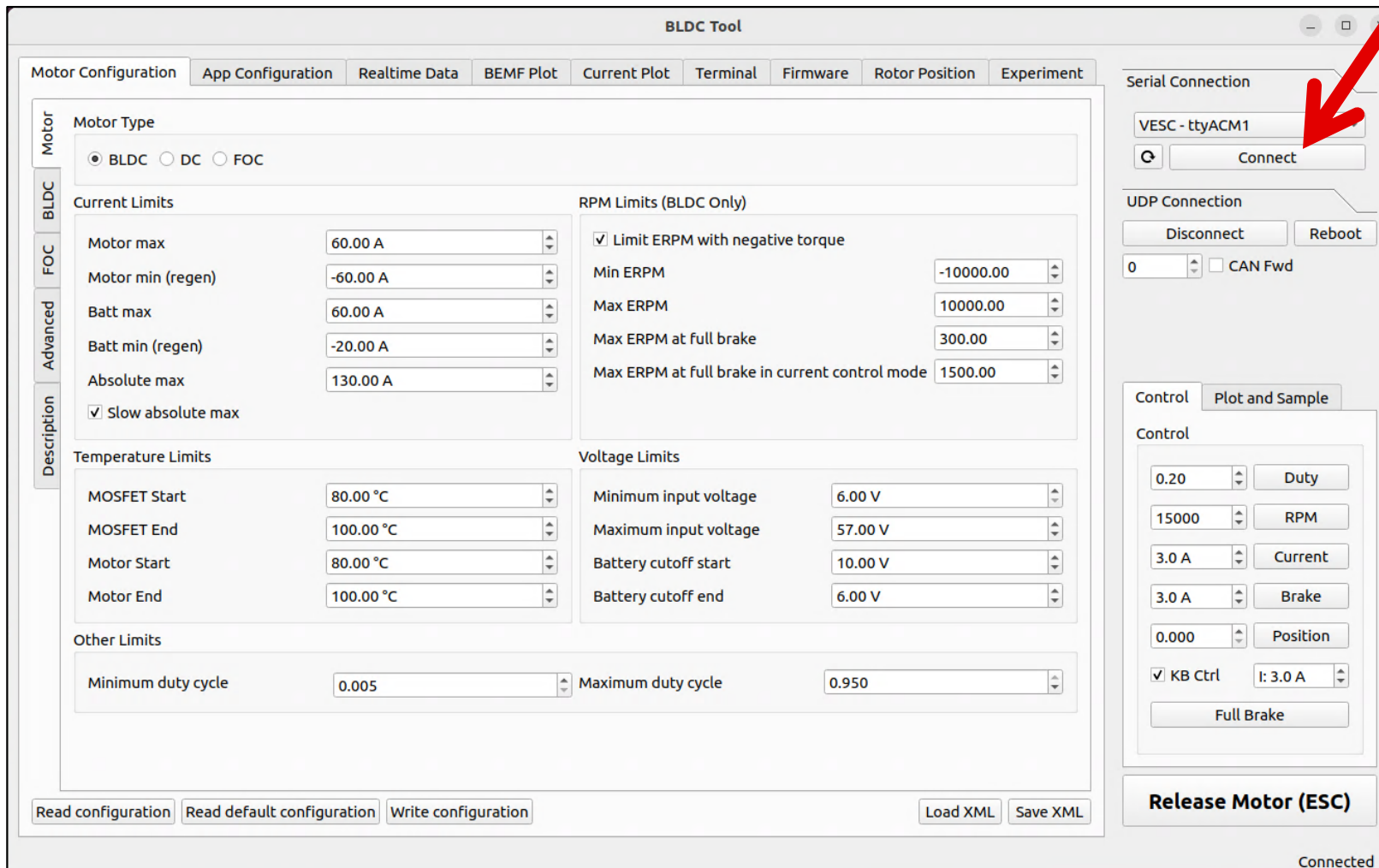
The screenshot shows the BLDC Tool interface with the following sections:

- Tabs:** Motor Configuration, App Configuration, Realtime Data, BEMF Plot, Current Plot, Terminal, Firmware, Rotor Position, Experiment.
- Motor Configuration:**
 - Motor Type:** BLDC (selected), DC, FOC.
 - Current Limits:**
 - Motor max: 60.00 A
 - Motor min (regen): -60.00 A
 - Batt max: 60.00 A
 - Batt min (regen): -20.00 A
 - Absolute max: 130.00 A
 - ☒ Slow absolute max
 - RPM Limits (BLDC Only):**
 - ☒ Limit ERPM with negative torque
 - Min ERPM: -10000.00
 - Max ERPM: 10000.00
 - Max ERPM at full brake: 300.00
 - Max ERPM at full brake in current control mode: 1500.00
 - Temperature Limits:**
 - MOSFET Start: 80.00 °C
 - MOSFET End: 100.00 °C
 - Motor Start: 80.00 °C
 - Motor End: 100.00 °C
 - Voltage Limits:**
 - Minimum input voltage: 6.00 V
 - Maximum input voltage: 57.00 V
 - Battery cutoff start: 10.00 V
 - Battery cutoff end: 6.00 V
 - Other Limits:**
 - Minimum duty cycle: 0.005
 - Maximum duty cycle: 0.950
- Serial Connection:**
 - Device: VESC - ttyACM1
 - Buttons: Connect, Disconnect, Reboot
- UDP Connection:**
 - Port: 0
 - ☐ CAN Fwd
- Control:**
 - Buttons: Plot and Sample, Control
 - Control settings:
 - Duty: 0.20
 - RPM: 15000
 - Current: 3.0 A
 - Brake: 3.0 A
 - Position: 0.000
 - ☒ KB Ctrl
 - I: 3.0 A
 - Full Brake button
 - Release Motor (ESC) button
- Buttons:** Read configuration, Read default configuration, Write configuration, Load XML, Save XML.
- Status:** Connected

VESC 설정값 세팅 : old_vesc_tool

- 아래와 같은 화면이 열리면 VESC와 연결한다.

(1) Connect 눌러서
VESC와 연결한다.



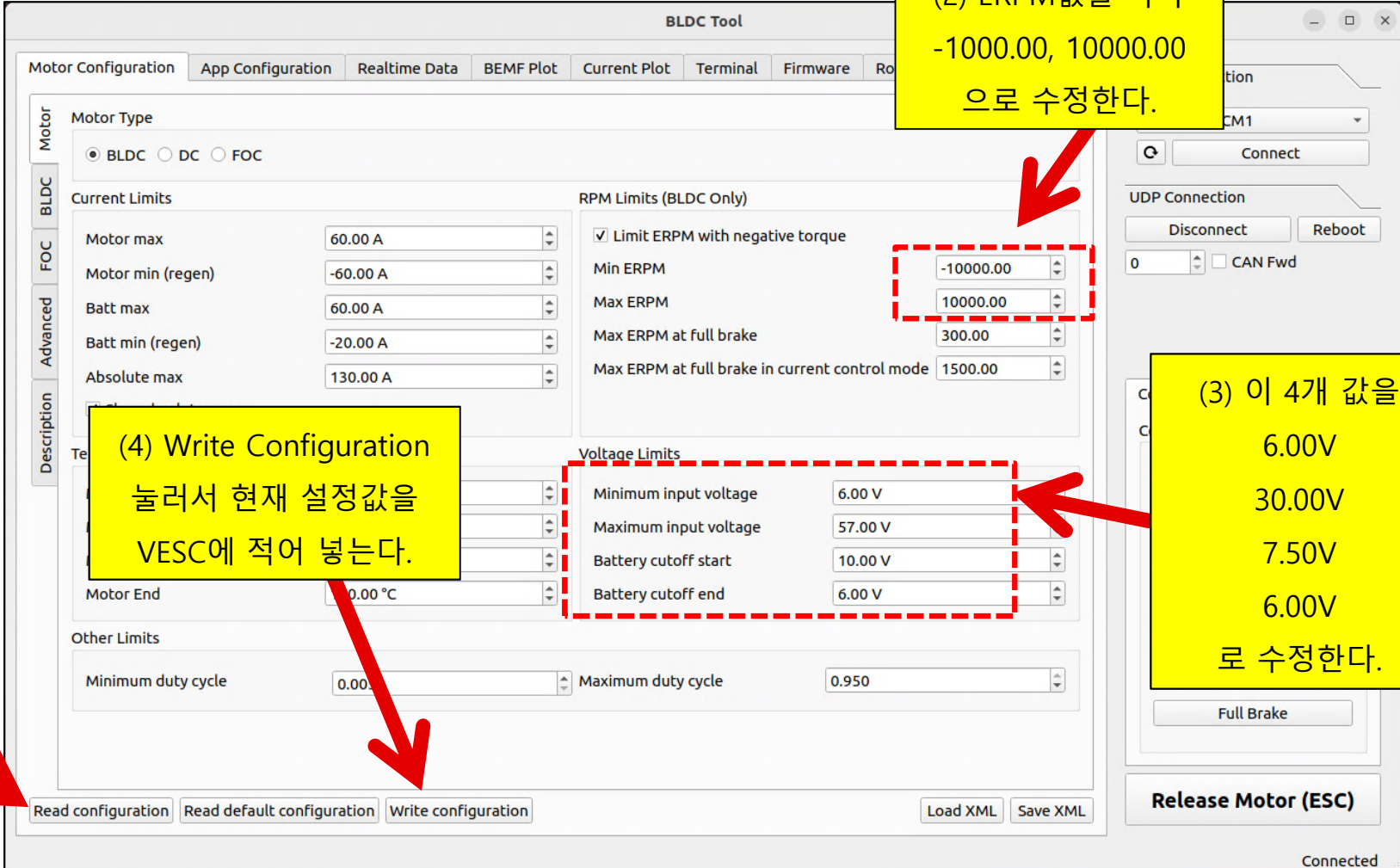
The screenshot shows the BLDC Tool interface with the following sections:

- Motor Configuration:**
 - Motor Type: ☒ BLDC, ☐ DC, ☐ FOC
 - Current Limits:
 - Motor max: 60.00 A
 - Motor min (regen): -60.00 A
 - Batt max: 60.00 A
 - Batt min (regen): -20.00 A
 - Absolute max: 130.00 A
 - ☒ Slow absolute max
 - RPM Limits (BLDC Only):
 - ☒ Limit ERPM with negative torque
 - Min ERPM: -10000.00
 - Max ERPM: 10000.00
 - Max ERPM at full brake: 300.00
 - Max ERPM at full brake in current control mode: 1500.00
 - Temperature Limits:
 - MOSFET Start: 80.00 °C
 - MOSFET End: 100.00 °C
 - Motor Start: 80.00 °C
 - Motor End: 100.00 °C
 - Voltage Limits:
 - Minimum input voltage: 6.00 V
 - Maximum input voltage: 57.00 V
 - Battery cutoff start: 10.00 V
 - Battery cutoff end: 6.00 V
 - Other Limits:
 - Minimum duty cycle: 0.005
 - Maximum duty cycle: 0.950
- Serial Connection:**
 - Device: VESC - ttyACM1
 - Buttons: Connect, Disconnect, Reboot
- UDP Connection:**
 - Port: 0
 - ☐ CAN Fwd
- Control:**
 - Buttons: Duty, RPM, Current, Brake, Position, Full Brake
 - ☒ KB Ctrl
 - Limit: I: 3.0 A
- Buttons at the bottom:** Read configuration, Read default configuration, Write configuration, Load XML, Save XML, Release Motor (ESC)
- Status:** Connected

(2) 연결 상태를
확인한다.

VEESC 설정값 세팅 : old_vesc_tool

- 아래 번호 (1)~(4) 순서대로 작업한다



The screenshot shows the BLDC Tool interface with the following configuration details:

- Motor Configuration:** Motor Type is BLDC.
- Current Limits:** Motor max (60.00 A), Motor min (regen) (-60.00 A), Batt max (60.00 A), Batt min (regen) (-20.00 A), Absolute max (130.00 A).
- RPM Limits (BLDC Only):** Limit ERPM with negative torque is checked. Min ERPM is -10000.00, Max ERPM is 10000.00, Max ERPM at full brake is 300.00, and Max ERPM at full brake in current control mode is 1500.00.
- Voltage Limits:** Minimum input voltage (6.00 V), Maximum input voltage (57.00 V), Battery cutoff start (10.00 V), and Battery cutoff end (6.00 V).
- Other Limits:** Minimum duty cycle (0.00) and Maximum duty cycle (0.950).

Annotations and actions:

- (1) Read Configuration**: 눌러서 현재 설정값을 읽어들인다. (Click to read current settings).
- (2) ERPM값을 각각 -10000.00, 10000.00 으로 수정한다.** (Modify ERPM values to -10000.00 and 10000.00).
- (3) 이 4개 값을 6.00V, 30.00V, 7.50V, 6.00V 로 수정한다.** (Modify these 4 values to 6.00V, 30.00V, 7.50V, 6.00V).
- (4) Write Configuration**: 눌러서 현재 설정값을 VESC에 적어 넣는다. (Click to write current settings to VESC).



끝